

Отчёт о лабораторной работе

Лабораторная работа 3

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1 Цель работы

Освоить работу с Wireshark: разобрать кадры Ethernet и проанализировать PDU транспортного и прикладного уровней стека TCP/IP.

2 Выполнение

Сначала выполним `ipconfig`. В выводе увидим несколько подключений, среди них и активное — беспроводная сеть. Из него мы узнаём `ipv6` и `ipv4` адреса, маску подсети и адрес основного шлюза (рис. 2.1).

```
Адаптер беспроводной локальной сети Подключение по локальной сети* 9:

Состояние среды. . . . . : Среда передачи недоступна.
DNS-суффикс подключения . . . . . :

Адаптер беспроводной локальной сети Подключение по локальной сети* 10:

Состояние среды. . . . . : Среда передачи недоступна.
DNS-суффикс подключения . . . . . :

Адаптер беспроводной локальной сети Беспроводная сеть:

DNS-суффикс подключения . . . . . :
IPv6-адрес. . . . . : fdd5:f158:1a24:0:4d0b:9f55:5f65:7986
Временный IPv6-адрес. . . . . : fdd5:f158:1a24:0:f937:8ce:945:21bd
Локальный IPv6-адрес канала . . . : fe80::e1b3:33f6:75fd:b4ae%23
IPv4-адрес. . . . . : 192.168.1.76
Маска подсети . . . . . : 255.255.255.0
Основной шлюз. . . . . : 192.168.1.1

Адаптер Ethernet Сетевое подключение Bluetooth:

Состояние среды. . . . . : Среда передачи недоступна.
DNS-суффикс подключения . . . . . :
PS C:\Users\dora2> |
```

Рис. 2.1: `ipconfig`

Если добавить к `ipconfig` ключ `/all`, данных станет больше — в частности, появится физический (mac) адрес. У нас он равен 44-a3-bb-57-9b-03. Из этих шести байт первые три — идентификатор производителя, оставшиеся три — самого интерфейса. Старший байт 44 говорит о том, что адрес индивидуальный и глобально администрируемый: два младших бита этого байта при переводе из hex в двоичную дают нули. Также в выводе видны `ipv4`-адрес 192.168.1.76, маска 255.255.255.0, шлюз 192.168.1.1, аренда получена 6 октября 2026 г. до 8:54:45

(рис. 2.2).

```
Адаптер беспроводной локальной сети Беспроводная сеть:
DNS-суффикс подключения . . . . . :
Описание. . . . . : Intel(R) Wi-Fi 6E AX211 160MHz
Физический адрес. . . . . : 44-A3-BB-57-9B-03
DHCP включен. . . . . : Да
Автонастройка включена. . . . . : Да
IPv6-адрес. . . . . : fdd5:f158:1a24:0:4d0b:9f55:5f65:7986(
Основной)
Временный IPv6-адрес. . . . . : fdd5:f158:1a24:0:f937:8ce:945:21bd(Ос
новной)
Локальный IPv6-адрес канала . . . : fe80::e1b3:33f6:75fd:b4ae%23(Основной
)
IPv4-адрес. . . . . : 192.168.1.76(Основной)
Маска подсети . . . . . : 255.255.255.0
Аренда получена. . . . . : 6 октября 2026 г. 1:54:45
Срок аренды истекает. . . . . : 6 октября 2026 г. 8:54:45
Основной шлюз. . . . . : 192.168.1.1
DHCP-сервер. . . . . : 192.168.1.1
IAID DHCPv6 . . . . . : 407151547
DUID клиента DHCPv6 . . . . . : 00-01-00-01-2F-47-EE-9A-34-5A-60-27-0
2-6B
DNS-серверы. . . . . : fe80::52ff:20ff:fedd:e4c6%23
                        77.88.8.8
                        77.88.8.1
                        fe80::52ff:20ff:fedd:e4c6%23
NetBios через TCP/IP. . . . . : Включен
```

Рис. 2.2: ipconfig /all

Открываем wireshark и выбираем активное сетевое подключение. У нас это беспроводная сеть (рис. 2.3).

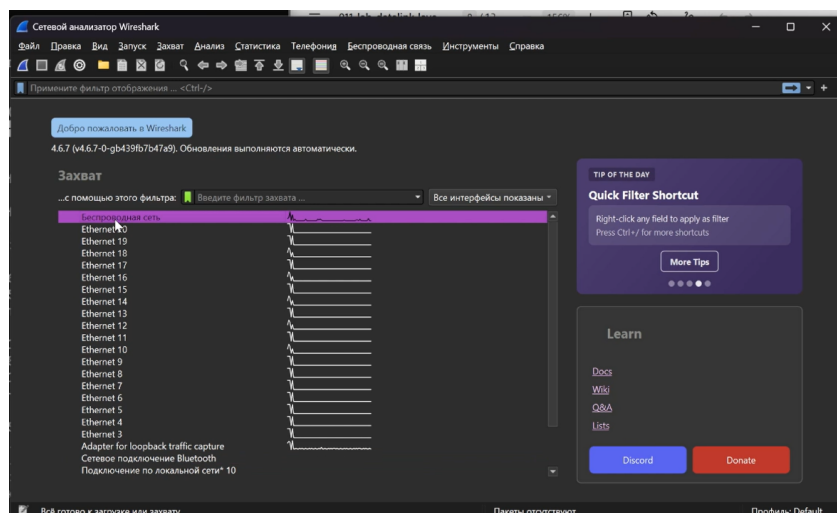


Рис. 2.3: Выбор соединения

Так выглядит сам интерфейс (рис. 2.4).

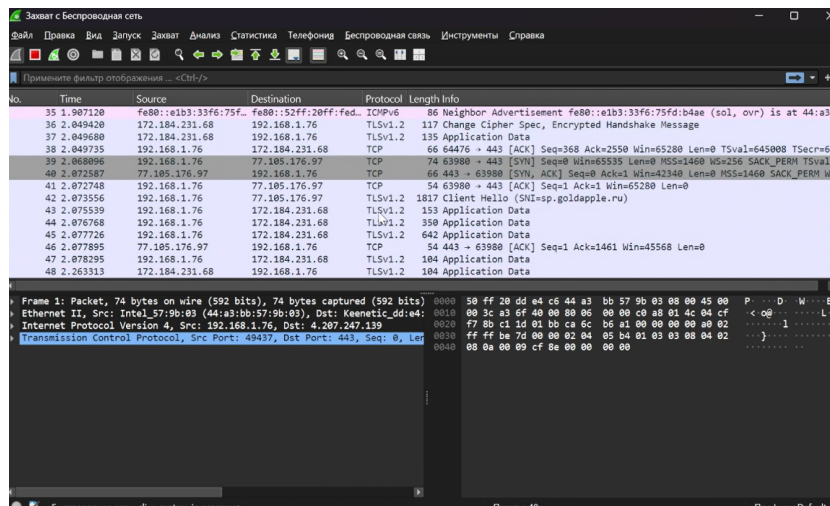


Рис. 2.4: Wireshark

Попробуем отправить ping на шлюз 192.168.1.1. Все 4 пакета получили ответ с TTL=64 и временем 1 мс. Потерь нет (рис. 2.5).

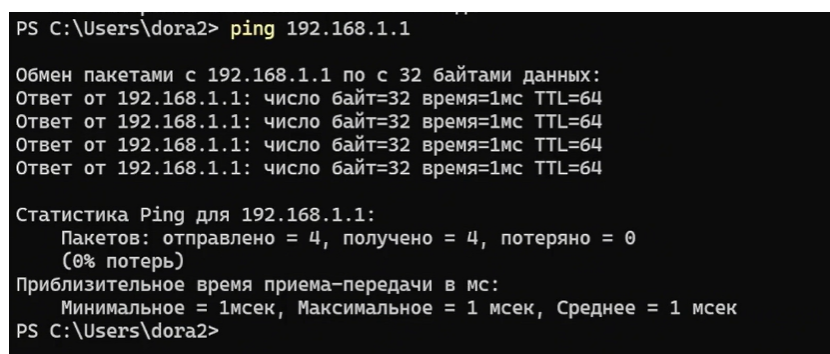


Рис. 2.5: ping

Возвращаемся в wireshark и смотрим на icmp-пакеты. Поскольку ушло 4 пакета, видим 8 строк — по одной на отправку и получение для каждого пакета. Отправитель — 192.168.1.76, получатель — 192.168.1.1 (рис. 2.6).

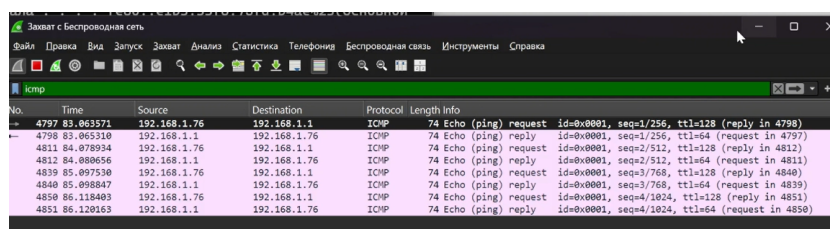


Рис. 2.6: icmp

Разберём самый первый, исходящий пакет. Длина кадра — 74 байта, тип — ethernet 2; физический адрес отправителя (наш) — Intel_57:9b:03 (44:a3:bb:57:9b:03), получателя (шлюз) — Keentic_dd:e4:c6 (50:ff:20:dd:e4:c6). Оба адреса индивидуальные и глобально администрируемые. Внутри — IPv4 и ICMP (рис. 2.7).

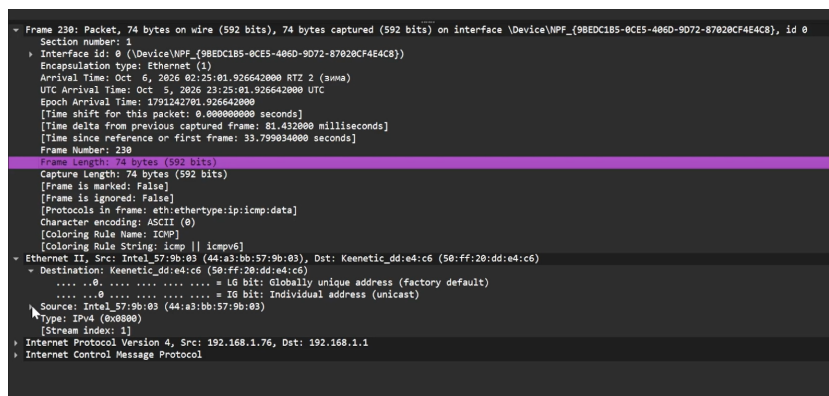


Рис. 2.7: Анализ исходящего пакета

Теперь посмотрим на ответный фрейм. Он почти не отличается от исходящего — только адреса отправителя и получателя стоят наоборот: 50:ff:20:dd:e4:c6 → 44:a3:bb:57:9b:03 (рис. 2.8).

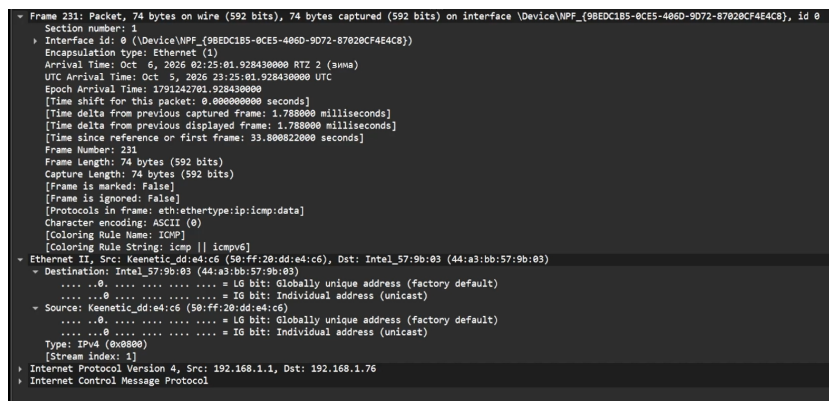


Рис. 2.8: Анализ входящего пакета

Переходим к арг-пакетам. Информации в них немного; задача этих пакетов — узнать физический адрес. Первый арг-пакет — запрос: Who has 192.168.1.76?

Tell 192.168.1.1, то есть шлюз пытается определить адрес нашего устройства (рис. 2.9).

No.	Time	Source	Destination	Protocol	Length	Info
229	33.717602	Intel_57:9b:03	Keenetic_dd:e4:c6	ARP	42	192.168.1.76 is at 44:a3:bb:57:9b:03
52	5.942725	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.149? Tell 192.168.1.1
65	7.884865	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.43? Tell 192.168.1.1
164	27.340617	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
166	28.373390	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.69? Tell 192.168.1.1
167	28.373390	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
168	28.373390	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.67? Tell 192.168.1.1
185	29.286850	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
186	29.389388	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.67? Tell 192.168.1.1
196	30.312166	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
197	30.413803	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.67? Tell 192.168.1.1
204	31.335563	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.67? Tell 192.168.1.1
228	33.717586	Keenetic_dd:e4:c6	Intel_57:9b:03	ARP	42	Who has 192.168.1.76? Tell 192.168.1.1

```

Frame 228: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{9BEDC1B5-0CE5-406D-9D72-87028CF4E4C8}, id 0
  Section number: 1
    Interface id: 0 (\Device\NPF_{9BEDC1B5-0CE5-406D-9D72-87028CF4E4C8})
    Encapsulation type: Ethernet (1)
    Arrival Time: Oct 5, 2026 02:25:01.845194000 RTZ 2 (sma)
    UTC Arrival Time: Oct 5, 2026 23:25:01.845194000 UTC
    Epoch Arrival Time: 1791242701.845194000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 92.409000 milliseconds]
    [Time delta from previous displayed frame: 2.38203000 seconds]
    [Time since reference or first frame: 33.717586000 seconds]
    Frame Number: 228
    Frame Length: 42 bytes (336 bits)
    Capture Length: 42 bytes (336 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:arp]
    Character encoding: ASCII (0)
    [Coloring Rule Name: ARP]
    [Coloring Rule String: arp]
    Ethernet II, Src: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6), Dst: Intel_57:9b:03 (44:a3:bb:57:9b:03)
      Destination: Intel_57:9b:03 (44:a3:bb:57:9b:03)
        ....0. .... = LG bit: Globally unique address (factory default)
        ....0. .... = IG bit: Individual address (unicast)
      Source: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6)
        ....0. .... = LG bit: Globally unique address (factory default)
        ....0. .... = IG bit: Individual address (unicast)
    Type: ARP (0x0806)
    [Stream index: 1]
    Address Resolution Protocol (request)
  
```

Рис. 2.9: arp

Второй arp-пакет — ответ: 192.168.1.76 is at 44:a3:bb:57:9b:03. В обоих случаях в заголовке ethernet 2 лежит физический адрес обоих устройств, но в первом пакете тип — arp (request), во втором — arp (reply) (рис. 2.10).

No.	Time	Source	Destination	Protocol	Length	Info
229	33.717602	Intel_57:9b:03	Keenetic_dd:e4:c6	ARP	42	192.168.1.76 is at 44:a3:bb:57:9b:03
52	5.942725	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.149? Tell 192.168.1.1
65	7.884865	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.43? Tell 192.168.1.1
164	27.340617	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
166	28.373390	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.69? Tell 192.168.1.1
167	28.373390	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
168	28.373390	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.67? Tell 192.168.1.1
185	29.286850	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
186	29.389388	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.67? Tell 192.168.1.1
196	30.312166	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
197	30.413803	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.67? Tell 192.168.1.1
204	31.335563	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.67? Tell 192.168.1.1
228	33.717586	Keenetic_dd:e4:c6	Intel_57:9b:03	ARP	42	Who has 192.168.1.76? Tell 192.168.1.1

```

Frame 229: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{9BEDC1B5-0CE5-406D-9D72-87028CF4E4C8}, id 0
  Section number: 1
    Interface id: 0 (\Device\NPF_{9BEDC1B5-0CE5-406D-9D72-87028CF4E4C8})
    Encapsulation type: Ethernet (1)
    Arrival Time: Oct 6, 2026 02:25:01.845210000 RTZ 2 (sma)
    UTC Arrival Time: Oct 5, 2026 23:25:01.845210000 UTC
    Epoch Arrival Time: 1791242701.845210000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 16.000 microseconds]
    [Time delta from previous displayed frame: 16.000 microseconds]
    [Time since reference or first frame: 33.717602000 seconds]
    Frame Number: 229
    Frame Length: 42 bytes (336 bits)
    Capture Length: 42 bytes (336 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:arp]
    Character encoding: ASCII (0)
    [Coloring Rule Name: ARP]
    [Coloring Rule String: arp]
    Ethernet II, Src: Intel_57:9b:03 (44:a3:bb:57:9b:03), Dst: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6)
      Destination: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6)
        ....0. .... = LG bit: Globally unique address (factory default)
        ....0. .... = IG bit: Individual address (unicast)
      Source: Intel_57:9b:03 (44:a3:bb:57:9b:03)
        ....0. .... = LG bit: Globally unique address (factory default)
        ....0. .... = IG bit: Individual address (unicast)
    Type: ARP (0x0806)
    [Stream index: 1]
    Address Resolution Protocol (reply)
  
```

Рис. 2.10: Второй arp пакет

Теперь пингуем rudn.ru — он не отвечает: превышен интервал ожидания для за-

проса, 4 отправлено, 0 получено, 100 % потерь. Зато ya.ru (77.88.44.242) отвечает: 4 отправлено, 4 получено (рис. 2.11).

```
PS C:\Users\dora2> ping rudn.ru

Обмен пакетами с rudn.ru [37.18.93.135] с 32 байтами данных:
Превышен интервал ожидания для запроса.
Превышен интервал ожидания для запроса.
Превышен интервал ожидания для запроса.
Превышен интервал ожидания для запроса.

Статистика Ping для 37.18.93.135:
  Пакетов: отправлено = 4, получено = 0, потеряно = 4
  (100% потерь)
PS C:\Users\dora2> ping ya.ru

Обмен пакетами с ya.ru [77.88.44.242] с 32 байтами данных:
Ответ от 77.88.44.242: число байт=32 время=5мс TTL=50
Ответ от 77.88.44.242: число байт=32 время=5мс TTL=50
Ответ от 77.88.44.242: число байт=32 время=5мс TTL=50
Ответ от 77.88.44.242: число байт=32 время=5мс TTL=50

Статистика Ping для 77.88.44.242:
  Пакетов: отправлено = 4, получено = 4, потеряно = 0
  (0% потерь)
Приблизительное время приема-передачи в мс:
  Минимальное = 5мсек, Максимальное = 5 мсек, Среднее = 5 мсек
PS C:\Users\dora2>
```

Рис. 2.11: ping rudn.ru и ya.ru

Смотрим на arp-пакеты для ya.ru: снова запрос Who has 192.168.1.76? Tell 192.168.1.1 и ответ 192.168.1.76 is at 44:a3:bb:57:9b:03 (рис. 2.12).

```
№  Time      Source      Destination  Protocol Length Info
18 5.210976  Keenetic_d... Keenetic_d... Broadcast    ARP        42 Who has 192.168.1.67? Tell 192.168.1.1
326 17.531932 Keenetic_d... Intel_57:9b:83 Broadcast    ARP        42 Who has 192.168.1.76? Tell 192.168.1.1
327 17.531971 Intel_57:9b:83 Keenetic_d... ARP        42 192.168.1.76 is at 44:a3:bb:57:9b:03
484 18.397837 Keenetic_d... Keenetic_d... Broadcast    ARP        42 Who has 192.168.1.73? Tell 192.168.1.1
496 19.311313 Keenetic_d... Keenetic_d... Broadcast    ARP        42 Who has 192.168.1.73? Tell 192.168.1.1
583 40.343381 Keenetic_d... Keenetic_d... Broadcast    ARP        42 Who has 192.168.1.69? Tell 192.168.1.1
583 40.437779 Keenetic_d... Keenetic_d... Broadcast    ARP        42 Who has 192.168.1.73? Tell 192.168.1.1
514 41.256987 Keenetic_d... Keenetic_d... Broadcast    ARP        42 Who has 192.168.1.73? Tell 192.168.1.1
523 44.413148 Keenetic_d... Intel_57:9b:83 ARP        42 Who has 192.168.1.76? Tell 192.168.1.1
524 44.413198 Intel_57:9b:83 Keenetic_d... ARP        42 192.168.1.76 is at 44:a3:bb:57:9b:03

Frame 326: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{98EDC185-0CE5-406D-9072-87020CF4E4C8}, id 0
  Section number: 1
  Interface id: 0 (\Device\NPF_{98EDC185-0CE5-406D-9072-87020CF4E4C8})
  Encapsulation type: Ethernet (1)
  Arrival Time: Oct 6, 2020 02:30:17.237678000 RT2.2 (sma)
  UTC Arrival Time: Oct 5, 2020 23:30:17.237678000 UTC
  Epoch Arrival Time: 1791243017.237678000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 179.654000 milliseconds]
  [Time delta from previous displayed frame: 11.328956000 seconds]
  [Time since reference or first frame: 17.531932000 seconds]
  Frame Number: 326
  Frame Length: 42 bytes (336 bits)
  Capture Length: 42 bytes (336 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: ethertype:arp]
  Character encoding: ASCII (0)
  [Coloring Rule Name: arp]
  [Coloring Rule String: arp]
  Ethernet II, Src: Keenetic_d... (30:ff:2b:dd:e4:c6), Dst: Intel_57:9b:83 (44:a3:bb:57:9b:03)
    Destination: Intel_57:9b:83 (44:a3:bb:57:9b:03)
      ....0. .... = LG bit: Globally unique address (factory default)
      ....0. .... = IG bit: Individual address (unicast)
    Source: Keenetic_d... (30:ff:2b:dd:e4:c6)
      ....0. .... = LG bit: Globally unique address (factory default)
      ....0. .... = IG bit: Individual address (unicast)
    Type: ARP (0x806)
    [Stream index: 0]
  Address Resolution Protocol (request)
```

Рис. 2.12: arp пакеты

Второй arp-пакет, как обычно, отличается только направлением (рис. 2.13).

No.	Time	Source	Destination	Protocol	Length	Info
18	5.210976	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.67? Tell 192.168.1.1
326	17.531932	Keenetic_dd:e4:c6	Intel_57:9b:03	ARP	42	Who has 192.168.1.76? Tell 192.168.1.1
327	17.531974	Intel_57:9b:03	Keenetic_dd:e4:c6	ARP	42	192.168.1.76 is at 44:a3:bb:57:9b:03
484	38.397837	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
496	39.311313	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
502	40.343381	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.69? Tell 192.168.1.1
503	40.437779	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
514	41.256987	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
523	44.413148	Keenetic_dd:e4:c6	Intel_57:9b:03	ARP	42	Who has 192.168.1.76? Tell 192.168.1.1
524	44.413198	Intel_57:9b:03	Keenetic_dd:e4:c6	ARP	42	192.168.1.76 is at 44:a3:bb:57:9b:03

▼ Frame 327: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{98EDC1B5-0CE5-486D-9072-87828CF4E4C8}, id 0

Section number: 1

Interface id: 0 (\Device\NPF_{98EDC1B5-0CE5-486D-9072-87828CF4E4C8})

Encapsulation type: Ethernet (1)

Arrival Time: Oct 6, 2026 02:30:17.237717000 RT2 2 (sma)

UTC Arrival Time: Oct 5, 2026 23:30:17.237717000 UTC

Epoch Arrival Time: 1791243017.237717000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 39.800 microseconds]

[Time delta from previous displayed frame: 39.800 microseconds]

[Time since reference on first frame: 17.531971000 seconds]

Frame Number: 327

Frame Length: 42 bytes (336 bits)

Capture Length: 42 bytes (336 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: ethertype:arp]

Character encoding: ASCII (0)

[Coloring Rule Name: ARP]

[Coloring Rule String: arp]

▼ Ethernet II, Src: Intel_57:9b:03 (44:a3:bb:57:9b:03), Dst: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6)

Destination: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6)

.....0..... = LG bit: Globally unique address (factory default)

.....0..... = IG bit: Individual address (unicast)

Source: Intel_57:9b:03 (44:a3:bb:57:9b:03)

.....0..... = LG bit: Globally unique address (factory default)

.....0..... = IG bit: Individual address (unicast)

Type: ARP (0x0806)

[Stream index: 0]

▼ Address Resolution Protocol (reply)

Рис. 2.13: Второй arp пакет

Теперь смотрим icmp-пакеты для ya.ru: их восемь. Обращаемся мы уже не к шлюзу, а напрямую к серверу Яндекса — ip-адрес 77.88.44.242. Однако на уровне ethernet 2 мы по-прежнему общаемся со шлюзом (рис. 2.14).

No.	Time	Source	Destination	Protocol	Length	Info
346	23.983268	192.168.1.76	37.18.93.135	ICMP	74	Echo (ping) request id=0x0001, seq=21/5376, ttl=128 (no response found!)
371	28.614618	192.168.1.76	37.18.93.135	ICMP	74	Echo (ping) request id=0x0001, seq=22/5632, ttl=128 (no response found!)
444	33.616253	192.168.1.76	37.18.93.135	ICMP	74	Echo (ping) request id=0x0001, seq=23/5888, ttl=128 (no response found!)
491	38.619478	192.168.1.76	37.18.93.135	ICMP	74	Echo (ping) request id=0x0001, seq=24/6144, ttl=128 (no response found!)
561	50.050613	192.168.1.76	77.88.44.242	ICMP	74	Echo (ping) request id=0x0001, seq=25/6400, ttl=128 (reply in 561)
562	50.055377	77.88.44.242	192.168.1.76	ICMP	74	Echo (ping) reply id=0x0001, seq=25/6400, ttl=50 (request in 561)
567	51.062456	192.168.1.76	77.88.44.242	ICMP	74	Echo (ping) request id=0x0001, seq=26/6656, ttl=128 (reply in 568)
568	51.067700	77.88.44.242	192.168.1.76	ICMP	74	Echo (ping) reply id=0x0001, seq=26/6656, ttl=50 (request in 567)
570	52.080488	192.168.1.76	77.88.44.242	ICMP	74	Echo (ping) request id=0x0001, seq=27/6912, ttl=128 (reply in 572)
572	52.085955	77.88.44.242	192.168.1.76	ICMP	74	Echo (ping) reply id=0x0001, seq=27/6912, ttl=50 (request in 570)
575	53.085587	192.168.1.76	77.88.44.242	ICMP	74	Echo (ping) request id=0x0001, seq=28/7168, ttl=128 (reply in 576)
576	53.091008	77.88.44.242	192.168.1.76	ICMP	74	Echo (ping) reply id=0x0001, seq=28/7168, ttl=50 (request in 575)

▼ Frame 561: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{98EDC1B5-0CE5-486D-9072-87828CF4E4C8}, id 0

Section number: 1

Interface id: 0 (\Device\NPF_{98EDC1B5-0CE5-486D-9072-87828CF4E4C8})

Encapsulation type: Ethernet (1)

Arrival Time: Oct 6, 2026 02:30:49.755859000 RT2 2 (sma)

UTC Arrival Time: Oct 5, 2026 23:30:49.755859000 UTC

Epoch Arrival Time: 1791243049.755859000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 3.162000 milliseconds]

[Time delta from previous displayed frame: 21.430635000 seconds]

[Time since reference on first frame: 50.050113000 seconds]

Frame Number: 561

Frame Length: 74 bytes (592 bits)

Capture Length: 74 bytes (592 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: ethertype:ip:icmp:data]

Character encoding: ASCII (0)

[Coloring Rule Name: ICMP]

[Coloring Rule String: icmp || icmpv6]

▼ Ethernet II, Src: Intel_57:9b:03 (44:a3:bb:57:9b:03), Dst: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6)

Destination: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6)

.....0..... = LG bit: Globally unique address (factory default)

.....0..... = IG bit: Individual address (unicast)

Source: Intel_57:9b:03 (44:a3:bb:57:9b:03)

.....0..... = LG bit: Globally unique address (factory default)

.....0..... = IG bit: Individual address (unicast)

Type: IPv4 (0x0800)

[Stream index: 0]

▼ Internet Protocol Version 4, Src: 192.168.1.76, Dst: 77.88.44.242

▼ Internet Control Message Protocol

Рис. 2.14: Исходящий пакет icmp

Входящий icmp — ответный. Отличается от исходного тем, что адреса поменяны местами: источник 77.88.44.242, получатель 192.168.1.76 (рис. 2.15).

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```

No.    Time           Source                Destination              Protocol Length Info
-----
364   14.933268      192.168.1.76             37.18.93.135            ICMP       74 Echo (ping) request  id=0x0001, seq=21/5376, ttl=128 (no response found)
371   28.614618      192.168.1.76             37.18.93.135            ICMP       74 Echo (ping) request  id=0x0001, seq=22/5632, ttl=128 (no response found)
441   33.616235      192.168.1.76             37.18.93.135            ICMP       74 Echo (ping) request  id=0x0001, seq=23/5888, ttl=128 (no response found)
494   38.619478      192.168.1.76             37.18.93.135            ICMP       74 Echo (ping) request  id=0x0001, seq=24/6144, ttl=128 (no response found)
573   50.810013      192.168.1.76             72.83.44.252            TCP        60 ECHO (ping) request     [seq=29409, win=65535, len=1] (reply=555)
> 562   50.855377      77.88.44.242             192.168.1.76           ICMP       74 Echo (ping) reply    id=0x0001, seq=25/6400, ttl=50 (request in 563)
> 562   51.062456      192.168.1.76             77.88.44.242            ICMP       74 Echo (ping) reply    id=0x0001, seq=26/6656, ttl=50 (reply in 568)
570   51.014000      192.168.1.76             192.168.1.76           ICMP       74 Echo (ping) request  id=0x0001, seq=27/6912, ttl=128 (request in 567)
570   52.880488      192.168.1.76             77.88.44.242            ICMP       74 Echo (ping) request  id=0x0001, seq=27/6912, ttl=128 (request in 572)
572   52.885955      192.168.1.76             192.168.1.76           ICMP       74 Echo (ping) reply    id=0x0001, seq=27/6912, ttl=50 (reply in 578)
573   53.050987      192.168.1.76             77.88.44.242            ICMP       74 Echo (ping) request  id=0x0001, seq=28/7168, ttl=128 (request in 576)
576   53.891080      77.88.44.242             192.168.1.76           ICMP       74 Echo (ping) reply    id=0x0001, seq=28/7168, ttl=50 (request in 575)

+ Frame 562: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface Device\NPF_{9BEC31B5-BCES-40E2-9D72-87828C7F4EC3}, Id 0
  Selection rules:
  + Interface id 0 (Device\NPF_{9BEC31B5-BCES-40E2-9D72-87828C7F4EC3})
    Encapsulation type: Ethernet (1)
  Arrival Time: Oct 6, 2026 82:30:49.71123800 RTT 2 (suma)
  UTC Arrival Time: Oct 5, 2026 23:30:49.761123800 UTC
  Epoch Arrival Time: 1791243849.761123800
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 3.264000 milliseconds]
[Time delta from previous displayed frame: 3.264000 milliseconds]
[Time since reference or first frame: 50.055377000 seconds]
Frame Number: 562
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: ethertype:ip:icmp:data]
Character encoding: ASCII (0)
[Coloring Rule Name: ICMP]
[Coloring Rule String: icmp ! icmpv6]
+ Ethernet II, Src: Keemetic-id:e4:c6 (50:f7:20:d4:e4:c6), Dst: Intel_57:9b:03 (44:a3:b7:5f:9b:03)
+ Destination: Intel_57:9b:03 (44:a3:b7:5f:9b:03)
.....0..... : IO bit: Globally unique address (factory default)
.....0..... : IO bit: Individual address (unicast)
+ Source: Keemetic-id:e4:c6 (50:f7:20:d4:e4:c6)
.....0..... : IO bit: Globally unique address (factory default)
.....0..... : IO bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream Index: 0]
+ Internet Protocol Version 4, Src: 77.88.44.242, Dst: 192.168.1.76
+ Internet Control Message Protocol

```

Откроем сайт по протоколу `http` — `info.cern.ch` (рис. 2.16).

Рис. 2.16: Подключение по http

Смотрим http-пакеты: GET /user/a03/www/default/NikhefGuide.html HTTP/1.1 уходит на 192.16.186.153 с порта 52037 на 80. Работает http поверх tcp — видим порты, длину сегмента, флаги (PSH, ACK) и чексумму (рис. 2.17).

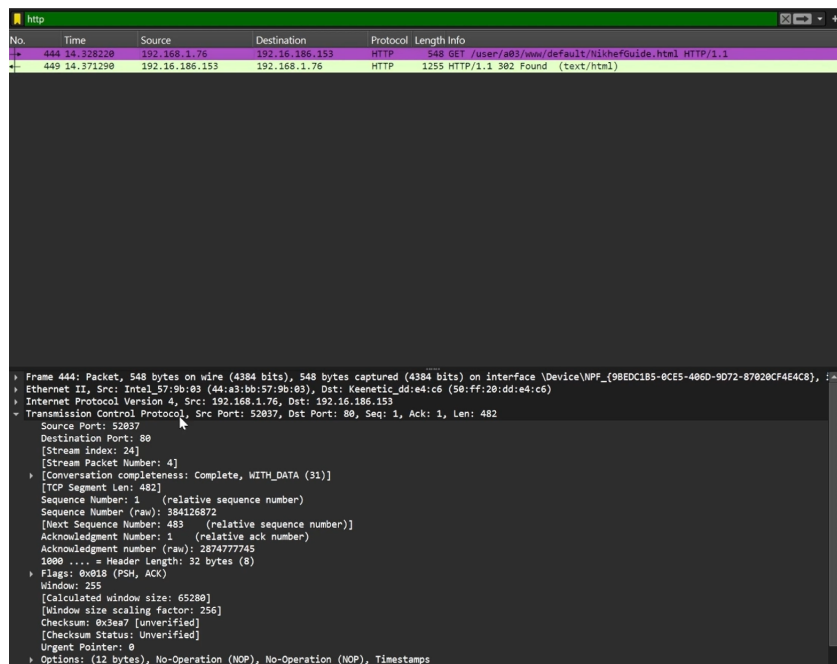


Рис. 2.17: Исходящий http пакет

У входящего пакета источник — 192.16.186.153, порт 80, а получатель — 192.168.1.76, порт 52037. Длина сегмента 1189, ответ HTTP/1.1 302 Found (рис. 2.18).

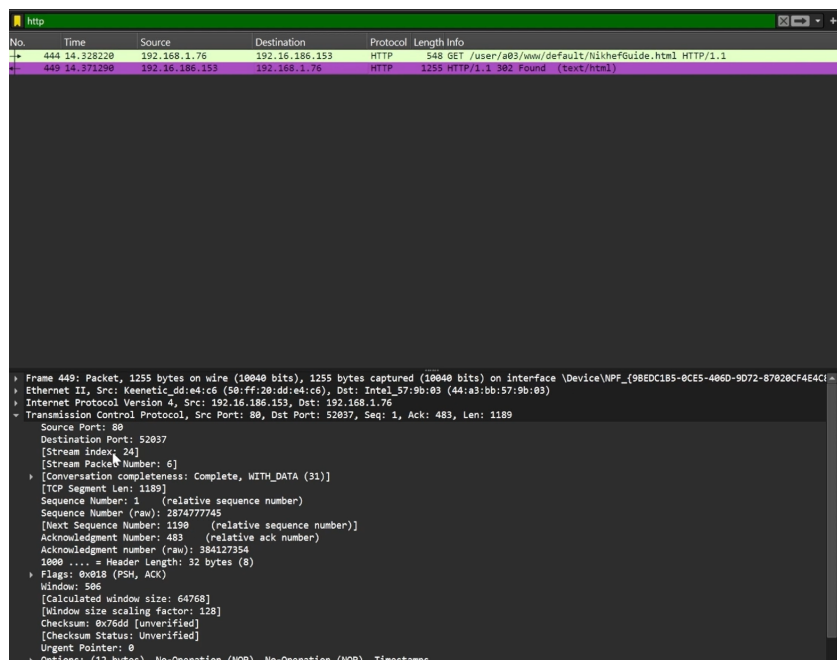


Рис. 2.18: Входящий http пакет

Теперь рассмотрим dns-пакеты. Они работают по udp, порт источника 53 — то есть пакеты приходят от DNS-сервера 77.88.8.1. Видны порты, длина и чексумма (рис. 2.19).

The screenshot shows the Wireshark interface with a list of network packets. The selected packet is number 457, which is a DNS response from 77.88.8.1 to 192.168.1.76. The detailed view below the packet list shows the structure of the DNS response, including the source and destination ports (53 and 49664), the length (129), the checksum (0x5e00), and the domain name system response.

No.	Time	Source	Destination	Protocol	Length	Info
408	13.883109	192.168.1.76	77.88.8.1	DNS	73	Standard query 0x10a4 HTTPS nic.nikhef.nl
409	13.883591	192.168.1.76	77.88.8.1	DNS	73	Standard query 0x10a4 A nic.nikhef.nl
410	13.995083	192.168.1.76	77.88.8.1	DNS	73	Standard query 0xf083 HTTPS nic.nikhef.nl
411	13.995371	192.168.1.76	77.88.8.1	DNS	73	Standard query 0x453b A nic.nikhef.nl
412	14.000331	192.168.1.76	77.88.8.1	DNS	73	Standard query 0xaa91 HTTPS nic.nikhef.nl
413	14.000584	192.168.1.76	77.88.8.1	DNS	73	Standard query 0x9e17 A nic.nikhef.nl
419	14.092082	77.88.8.1	192.168.1.76	DNS	131	Standard query response 0x10a4 A nic.nikhef.nl CNAME www.nikhef.nl
420	14.110093	77.88.8.1	192.168.1.76	DNS	181	Standard query response 0x453b A nic.nikhef.nl CNAME www.nikhef.nl
424	14.189724	77.88.8.1	192.168.1.76	DNS	131	Standard query response 0x9e17 A nic.nikhef.nl CNAME www.nikhef.nl
425	14.201846	77.88.8.1	192.168.1.76	DNS	131	Standard query response 0x9e17 A nic.nikhef.nl CNAME www.nikhef.nl
426	14.210088	77.88.8.1	192.168.1.76	DNS	181	Standard query response 0xf083 HTTPS nic.nikhef.nl CNAME www.nikhef.nl
429	14.223595	77.88.8.1	192.168.1.76	DNS	181	Standard query response 0xaa91 HTTPS nic.nikhef.nl CNAME www.nikhef.nl
450	14.373630	192.168.1.76	77.88.8.1	DNS	73	Standard query 0xdf80 HTTPS www.nikhef.nl
451	14.373765	192.168.1.76	77.88.8.1	DNS	73	Standard query 0xa274 A www.nikhef.nl
453	14.506532	77.88.8.1	192.168.1.76	DNS	113	Standard query response 0xa274 A www.nikhef.nl CNAME lvp029web.nikhef.nl
457	14.516570	77.88.8.1	192.168.1.76	DNS	163	Standard query response 0xdf80 HTTPS www.nikhef.nl CNAME lvp029web.nikhef.nl
519	15.044930	192.168.1.76	77.88.8.1	DNS	89	Standard query 0x0501 A ya.ru
527	15.045488	192.168.1.76	77.88.8.1	DNS	112	Standard query 0xed6c A browser.translate.yandex.net
529	15.045606	192.168.1.76	77.88.8.1	DNS	112	Standard query 0x5c55 HTTPS browser.translate.yandex.net
531	15.045715	192.168.1.76	77.88.8.1	DNS	89	Standard query 0xc928 HTTPS ya.ru
534	15.050697	77.88.8.1	192.168.1.76	DNS	139	Standard query response 0x0501 A ya.ru A 5.255.255.242 A 77.88.44.24
539	15.051745	77.88.8.1	192.168.1.76	DNS	130	Standard query response 0xed6c A browser.translate.yandex.net A 87.2
543	15.051745	77.88.8.1	192.168.1.76	DNS	165	Standard query response 0x5c55 HTTPS browser.translate.yandex.net 50
544	15.051745	77.88.8.1	192.168.1.76	DNS	147	Standard query response 0xc928 HTTPS ya.ru SOA ns1.yandex.ru
622	15.235544	192.168.1.76	77.88.8.1	DNS	105	Standard query 0x2008 HTTPS api.browser.yandex.ru

Frame 457: Packet, 163 bytes on wire (1304 bits), 163 bytes captured (1304 bits) on interface \Device\NPF_{9BEDC1B5-0CE5-4060-9072-87028CF4E4C8}, id
Ethernet II, Src: KaeNetic_dde4:c6 (58:ff:28:dd:e4:c6), Dst: Intel_57:9b:03 (44:a3:bb:57:9b:03)
Internet Protocol Version 4, Src: 77.88.8.1, Dst: 192.168.1.76
User Datagram Protocol, Src Port: 53, Dst Port: 49664
Source Port: 53
Destination Port: 49664
Length: 129
Checksum: 0x5e00 [unverified]
[Checksum Status: Unverified]
[Stream index: 8]
[Stream Packet Number: 2]
[Timestamps]
UDP payload (121 bytes)
Domain Name System (response)

Рис. 2.19: Исходящий dns пакет

Входящий dns-пакет на уровне udp отличается портами: источник 53, получатель 49664. Также отличаются длина и чексумма (рис. 2.20).

No.	Time	Source	Destination	Protocol	Length	Info
408	13.883109	192.168.1.76	77.88.8.1	DNS	73	Standard query 0x5c59 HTTPS nic.nikhef.nl
409	13.883591	192.168.1.76	77.88.8.1	DNS	73	Standard query 0x18a4 A nic.nikhef.nl
410	13.995083	192.168.1.76	77.88.8.1	DNS	73	Standard query 0xf883 HTTPS nic.nikhef.nl
411	13.995371	192.168.1.76	77.88.8.1	DNS	73	Standard query 0x53b0 A nic.nikhef.nl
412	14.000331	192.168.1.76	77.88.8.1	DNS	73	Standard query 0xaa91 HTTPS nic.nikhef.nl
413	14.000584	192.168.1.76	77.88.8.1	DNS	73	Standard query 0x9e17 A nic.nikhef.nl
419	14.092002	77.88.8.1	192.168.1.76	DNS	131	Standard query response 0x18a4 A nic.nikhef.nl CNAME www.nikhef.nl C
420	14.119093	77.88.8.1	192.168.1.76	DNS	181	Standard query response 0x5c59 HTTPS nic.nikhef.nl CNAME www.nikhef.
424	14.109724	77.88.8.1	192.168.1.76	DNS	131	Standard query response 0x53b0 A nic.nikhef.nl CNAME www.nikhef.nl C
425	14.201846	77.88.8.1	192.168.1.76	DNS	131	Standard query response 0x9e17 A nic.nikhef.nl CNAME www.nikhef.nl C
426	14.210008	77.88.8.1	192.168.1.76	DNS	181	Standard query response 0xf883 HTTPS nic.nikhef.nl CNAME www.nikhef.
429	14.223595	77.88.8.1	192.168.1.76	DNS	181	Standard query response 0xaa91 HTTPS nic.nikhef.nl CNAME www.nikhef.
450	14.373630	192.168.1.76	77.88.8.1	DNS	73	Standard query 0xf880 HTTPS www.nikhef.nl
451	14.373765	192.168.1.76	77.88.8.1	DNS	73	Standard query 0xa274 A www.nikhef.nl
453	14.506532	77.88.8.1	192.168.1.76	DNS	113	Standard query response 0xa274 A www.nikhef.nl CNAME lvp82web.nikhe
457	14.576970	77.88.8.1	192.168.1.76	DNS	163	Standard query response 0xf880 HTTPS www.nikhef.nl CNAME lvp82web.nikhe
519	15.044530	192.168.1.76	77.88.8.1	DNS	89	Standard query 0xc555 ya.ru
527	15.045488	192.168.1.76	77.88.8.1	DNS	112	Standard query 0xed6c A browser.translate.yandex.net
529	15.045606	192.168.1.76	77.88.8.1	DNS	112	Standard query 0xc555 HTTPS browser.translate.yandex.net
531	15.045715	192.168.1.76	77.88.8.1	DNS	89	Standard query 0xc528 HTTPS ya.ru
534	15.050657	77.88.8.1	192.168.1.76	DNS	139	Standard query response 0x8591 A ya.ru A 5.255.255.242 A 77.88.44.24
539	15.051745	77.88.8.1	192.168.1.76	DNS	130	Standard query response 0xed6c A browser.translate.yandex.net A 87.2
543	15.051745	77.88.8.1	192.168.1.76	DNS	165	Standard query response 0xc555 HTTPS browser.translate.yandex.net 50
544	15.051745	77.88.8.1	192.168.1.76	DNS	147	Standard query response 0xc528 HTTPS ya.ru SOA ns1.yandex.ru
622	15.235544	192.168.1.76	77.88.8.1	DNS	105	Standard query 0x2088 HTTPS api.browser.yandex.ru

Frame 519: Packet, 89 bytes on wire (712 bits), 89 bytes captured (712 bits) on interface \Device\NPF_{98EDC1B5-0CE5-406D-9072-87020CF4E4C8}, id 0
 Ethernet II, Src: Intel_57:9b:03 (44:a3:bb:57:9b:03), Dst: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6)
 Internet Protocol Version 4, Src: 192.168.1.76, Dst: 77.88.8.1
 Transmission Control Protocol, Src Port: 53821, Dst Port: 53, Seq: 3, Ack: 1, Len: 23
 Source Port: 53821
 Destination Port: 53
 [Stream index: 29]
 [Stream Packet Number: 5]
 [Conversation completeness: Complete, WITH_DATA (31)]
 [TCP Segment Len: 23]
 Sequence Number: 3 (relative sequence number)
 Sequence Number (raw): 335895115
 [Next Sequence Number: 26 (relative sequence number)]
 Acknowledgment Number: 1 (relative ack number)
 Acknowledgment number (raw): 748858784
 2000 = Header Length: 32 bytes (8)
 Flags: 0x018 (PSH, ACK)
 Window: 255
 [Calculated window size: 65280]
 [Window size scaling factor: 256]
 Checksum: 0x78b [unverified]
 [Checksum Status: Unverified]
 Urgent Pointer: 0
 Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
 [Timestamps]
 [SEQ/ACK analysis]
 [Client Contiguous Streams: 1]
 Domain Name System Protocol

Рис. 2.20: Входящий dns пакет

Теперь посмотрим на quic. Он работает поверх udp: порт источника 54581, получателя — 443. Внутри udp содержится примерно та же информация, что и в dns: порты, длина, чексумма (рис. 2.21).

No.	Time	Source	Destination	Protocol	Length	Info
8508	32.924096	192.168.1.76	172.217.119.4	QUIC	1292	Initial, DCID=ed70d1ed3bd0eec5, PKN: 1, CRYPTO
8509	32.924208	192.168.1.76	172.217.119.4	QUIC	1292	Initial, DCID=ed70d1ed3bd0eec5, PKN: 2, CRYPTO
8510	32.924264	192.168.1.76	172.217.119.4	QUIC	1292	Initial, DCID=ed70d1ed3bd0eec5, PKN: 3, CRYPTO, CRYPTO, PADDING, CRYPTO...
8511	32.924468	192.168.1.76	172.217.119.4	QUIC	121	0-RTT, DCID=ed70d1ed3bd0eec5
8512	32.948714	172.217.119.4	192.168.1.76	QUIC	82	Initial, SCID=ed70d1ed3bd0eec5, PKN: 1, ACK
8513	32.948841	172.217.119.4	192.168.1.76	QUIC	82	Initial, SCID=ed70d1ed3bd0eec5, PKN: 2, ACK
8514	32.949216	172.217.119.4	192.168.1.76	QUIC	1292	Initial, SCID=ed70d1ed3bd0eec5, PKN: 3, ACK, PADDING
8515	32.949433	172.217.119.4	192.168.1.76	QUIC	1292	Initial, SCID=ed70d1ed3bd0eec5, PKN: 4, ACK, PADDING
8516	32.949945	172.217.119.4	192.168.1.76	QUIC	1292	Protected Payload (KP0)
8517	32.949945	172.217.119.4	192.168.1.76	QUIC	857	Protected Payload (KP0)
8518	32.949945	172.217.119.4	192.168.1.76	QUIC	265	Protected Payload (KP0)
8519	32.950613	192.168.1.76	172.217.119.4	QUIC	136	Handshake, DCID=ed70d1ed3bd0eec5
8520	32.950842	192.168.1.76	172.217.119.4	QUIC	73	Protected Payload (KP0), DCID=ed70d1ed3bd0eec5
8521	32.951249	192.168.1.76	172.217.119.4	QUIC	1288	Protected Payload (KP0), DCID=ed70d1ed3bd0eec5
8522	32.951327	192.168.1.76	172.217.119.4	QUIC	298	Protected Payload (KP0), DCID=ed70d1ed3bd0eec5
8524	32.974163	172.217.119.4	192.168.1.76	QUIC	66	Protected Payload (KP0)
8525	32.974303	172.217.119.4	192.168.1.76	QUIC	162	Protected Payload (KP0)
8526	32.974395	172.217.119.4	192.168.1.76	QUIC	71	Protected Payload (KP0)
8527	32.974955	172.217.119.4	192.168.1.76	QUIC	67	Protected Payload (KP0)
8528	32.975405	192.168.1.76	172.217.119.4	QUIC	73	Protected Payload (KP0), DCID=ed70d1ed3bd0eec5
8533	33.425728	172.217.119.4	192.168.1.76	QUIC	1275	Protected Payload (KP0)
8536	33.453616	192.168.1.76	172.217.119.4	QUIC	78	Protected Payload (KP0), DCID=ed70d1ed3bd0eec5
8538	33.502758	172.217.119.4	192.168.1.76	QUIC	66	Protected Payload (KP0)

Frame 8528: Packet, 73 bytes on wire (584 bits), 73 bytes captured (584 bits) on interface \Device\NPF_{98EDC1B5-0CE5-406D-9072-87020CF4E4C8}, id 0
 Ethernet II, Src: Intel_57:9b:03 (44:a3:bb:57:9b:03), Dst: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6)
 Internet Protocol Version 4, Src: 192.168.1.76, Dst: 172.217.119.4
 User Datagram Protocol, Src Port: 54581, Dst Port: 443
 Source Port: 54581
 Destination Port: 443
 Length: 39
 Checksum: 0xe60a [unverified]
 [Checksum Status: Unverified]
 [Stream index: 46]
 [Stream Packet Number: 20]
 [Timestamps]
 UDP payload (31 bytes)
 QUIC IETF

Рис. 2.21: Исходящий quic пакет

У входящего quic-пакета порты идут в обратном порядке: 443 → 54581. Также отличаются длина и чексумма (рис. 2.22).

The screenshot shows a Wireshark capture of QUIC traffic. The top pane displays a list of packets, with packet 8527 highlighted in purple. The bottom pane shows the details of packet 8527, which is a QUIC packet of length 33 bytes, captured on interface \Device\NPF_{98E0C185-0CE5-4060-9072-87020CF4E4C8}, id 0. The packet is an Internet Protocol Version 4, Src: 172.217.119.4, Dst: 192.168.1.76. The User Datagram Protocol section shows Source Port: 443, Destination Port: 54581, Length: 33, Checksum: 0xe51d [unverified], [Checksum Status: Unverified], [Stream Index: 46], [Stream Packet Number: 19], [Timestamps], UDP payload (25 bytes), and QUIC IETF.

No.	Time	Source	Destination	Protocol	Length	Info
8508	32.924095	192.168.1.76	172.217.119.4	QUIC	1292	Initial, DCID=ed70d1ed3bd0eec5, PKN: 1, CRYPTO
8509	32.924208	192.168.1.76	172.217.119.4	QUIC	1292	Initial, DCID=ed70d1ed3bd0eec5, PKN: 2, CRYPTO
8510	32.924264	192.168.1.76	172.217.119.4	QUIC	1292	Initial, DCID=ed70d1ed3bd0eec5, PKN: 3, CRYPTO, CRYPTO, PADDING, CRYPTO...
8511	32.924468	192.168.1.76	172.217.119.4	QUIC	121	0-RTT, DCID=ed70d1ed3bd0eec5
8512	32.948714	172.217.119.4	192.168.1.76	QUIC	82	Initial, SCID=ed70d1ed3bd0eec5, PKN: 1, ACK
8513	32.948841	172.217.119.4	192.168.1.76	QUIC	82	Initial, SCID=ed70d1ed3bd0eec5, PKN: 2, ACK
8514	32.949216	172.217.119.4	192.168.1.76	QUIC	1292	Initial, SCID=ed70d1ed3bd0eec5, PKN: 3, ACK, PADDING
8515	32.949433	172.217.119.4	192.168.1.76	QUIC	1292	Initial, SCID=ed70d1ed3bd0eec5, PKN: 4, ACK, PADDING
8516	32.949945	172.217.119.4	192.168.1.76	QUIC	1292	Protected Payload (KPo)
8517	32.949945	172.217.119.4	192.168.1.76	QUIC	857	Protected Payload (KPo)
8518	32.949945	172.217.119.4	192.168.1.76	QUIC	265	Protected Payload (KPo)
8519	32.950613	192.168.1.76	172.217.119.4	QUIC	136	Handshake, DCID=ed70d1ed3bd0eec5
8520	32.950842	192.168.1.76	172.217.119.4	QUIC	73	Protected Payload (KPo), DCID=ed70d1ed3bd0eec5
8521	32.951249	192.168.1.76	172.217.119.4	QUIC	1288	Protected Payload (KPo), DCID=ed70d1ed3bd0eec5
8522	32.951327	192.168.1.76	172.217.119.4	QUIC	398	Protected Payload (KPo), DCID=ed70d1ed3bd0eec5
8524	32.974163	172.217.119.4	192.168.1.76	QUIC	66	Protected Payload (KPo)
8525	32.974383	172.217.119.4	192.168.1.76	QUIC	162	Protected Payload (KPo)
8526	32.974995	172.217.119.4	192.168.1.76	QUIC	71	Protected Payload (KPo)
8527	32.974995	172.217.119.4	192.168.1.76	QUIC	67	Protected Payload (KPo)
8528	32.975405	192.168.1.76	172.217.119.4	QUIC	73	Protected Payload (KPo), DCID=ed70d1ed3bd0eec5
8533	33.425728	172.217.119.4	192.168.1.76	QUIC	1275	Protected Payload (KPo)
8536	33.453616	192.168.1.76	172.217.119.4	QUIC	78	Protected Payload (KPo), DCID=ed70d1ed3bd0eec5
8538	33.502758	172.217.119.4	192.168.1.76	QUIC	66	Protected Payload (KPo)

Рис. 2.22: Входящий quic пакет

Теперь разберём процесс хендшейка. Для этого обновим http-страницу и посмотрим, что происходит перед http-запросом get. Сначала клиент (192.168.1.76, порт 57608) отправляет серверу (13.249.8.110, порт 443) syn-запрос. Во флагах tcp видим только SYN, sequence number — 0 (рис. 2.23).

No.	Time	Source	Destination	Protocol	Length	Info
207	4.164313	13.249.8.110	192.168.1.76	TLSv1.2	213	Server Hello, Change Cipher Spec, Encrypted Handshake Message
208	4.166121	192.168.1.76	13.249.8.110	TLSv1.2	249	Change Cipher Spec, Encrypted Handshake Message, Application Data
209	4.170487	192.168.1.76	5.255.255.242	TLSv1.2	1933	Application Data
210	4.170613	192.168.1.76	5.255.255.242	TLSv1.2	98	Application Data
211	4.176209	5.255.255.242	192.168.1.76	TCP	66	443 → 58203 [ACK] Seq=1781 Ack=5893 Win=166 Len=0 TSval=2878401396 TSecr=15
212	4.176691	5.255.255.242	192.168.1.76	TLSv1.2	101	Application Data
213	4.181512	5.255.255.242	192.168.1.76	TLSv1.2	244	Application Data
214	4.181567	192.168.1.76	5.255.255.242	TCP	66	58203 → 443 [ACK] Seq=5893 Ack=1994 Win=255 Len=0 TSval=3050196 TSecr=15
215	4.188402	13.249.8.110	192.168.1.76	TCP	66	443 → 57607 [ACK] Seq=148 Ack=527 Win=68096 Len=0 TSval=2123558872 TSecr=15
216	4.309262	13.249.8.110	192.168.1.76	TLSv1.2	429	Application Data
217	4.310849	192.168.1.76	13.249.8.110	TCP	66	57607 → 443 [FIN, ACK] Seq=527 Ack=511 Win=65024 Len=0 TSval=3050326 TSecr=15
218	4.312101	192.168.1.76	13.249.8.110	TCP	74	57608 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM TSval=2123558 TSecr=15
219	4.332555	13.249.8.110	192.168.1.76	TCP	66	443 → 57607 [FIN, ACK] Seq=511 Ack=528 Win=68096 Len=0 TSval=2123558 TSecr=15
220	4.332658	192.168.1.76	13.249.8.110	TCP	66	57607 → 443 [ACK] Seq=528 Ack=512 Win=65024 Len=0 TSval=3050347 TSecr=15
221	4.336276	13.249.8.110	192.168.1.76	TCP	74	443 → 57608 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 SACK_PERM TSval=2123558 TSecr=15
222	4.336441	192.168.1.76	13.249.8.110	TCP	66	57608 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0 TSval=3050351 TSecr=15
223	4.337092	192.168.1.76	13.249.8.110	TLSv1.2	409	Client Hello (SN=apps.overwolf.com)
224	4.361678	13.249.8.110	192.168.1.76	TCP	66	443 → 57608 [ACK] Seq=1 Ack=344 Win=67072 Len=0 TSval=1591387068 TSecr=15
225	4.361908	13.249.8.110	192.168.1.76	TLSv1.2	213	Server Hello, Change Cipher Spec, Encrypted Handshake Message
226	4.362787	192.168.1.76	13.249.8.110	TLSv1.2	222	Change Cipher Spec, Encrypted Handshake Message, Application Data
227	4.385518	13.249.8.110	192.168.1.76	TCP	66	443 → 57608 [ACK] Seq=148 Ack=500 Win=68096 Len=0 TSval=1591387084 TSecr=15
228	4.387193	13.249.8.110	192.168.1.76	TLSv1.2	667	Application Data
229	4.434332	192.168.1.76	13.249.8.110	TCP	66	57608 → 443 [ACK] Seq=500 Ack=749 Win=64768 Len=0 TSval=3050449 TSecr=15
230	5.611391	192.168.1.76	188.104.67.127	TCP	74	61705 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM TSval=2123558 TSecr=15
231	5.686990	192.168.1.76	5.255.255.242	TLSv1.2	437	Application Data


```

[Header checksum status: Unverified]
Source Address: 192.168.1.76
Destination Address: 13.249.8.110
[Stream index: 7]
Transmission Control Protocol, Src Port: 57608, Dst Port: 443, Seq: 0, Len: 0
Source Port: 57608
Destination Port: 443
[Stream index: 13]
[Stream Packet Number: 1]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 1792191646
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment Number (raw): 0
1010 .... = Header Length: 40 bytes (10)
Flags: 0x0002 (SYN)
Window: 65535
[Calculated window size: 65535]
Checksum: 0xd889 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (20 bytes), Maximum segment size, No-Operation (NOP), Window scale, SACK permitted, Timestamps
[Timestamps]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]

```

Рис. 2.23: syn

Сервер отвечает syn и ack: источник 13.249.8.110, порт 443, получатель 192.168.1.76, порт 57608. Во флагах видим SYN, ACK, sequence number 0, acknowledgment number 1 (рис. 2.24).

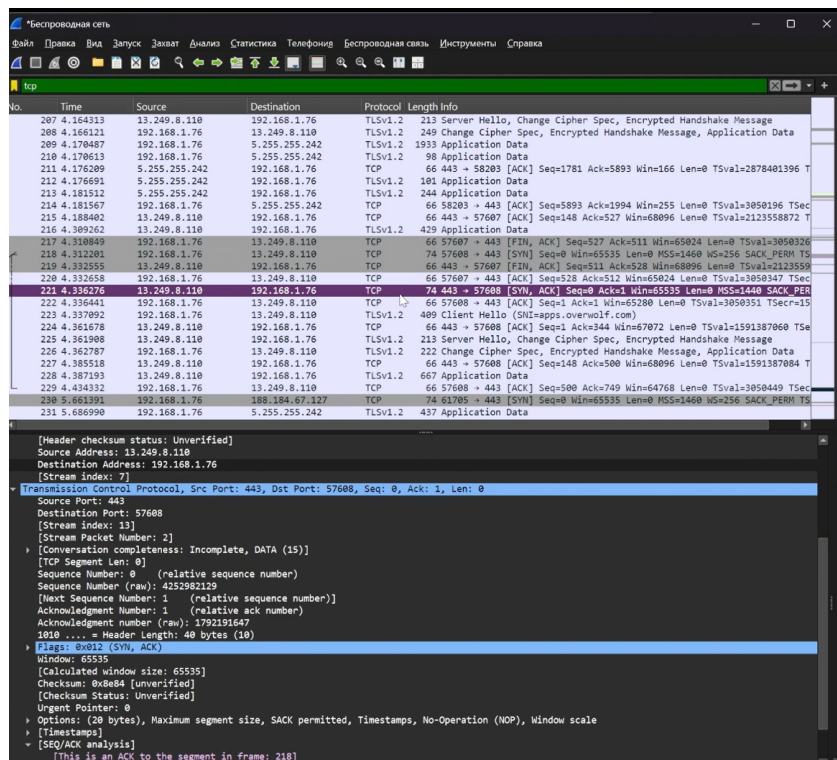


Рис. 2.24: syn + ack

Клиенту остаётся пожать руку — отправить ack. Источник 192.168.1.76, порт 57608, получатель 13.249.8.110, порт 443. Во флагах только ACK, sequence number 1, acknowledgment number 1 (рис. 2.25).

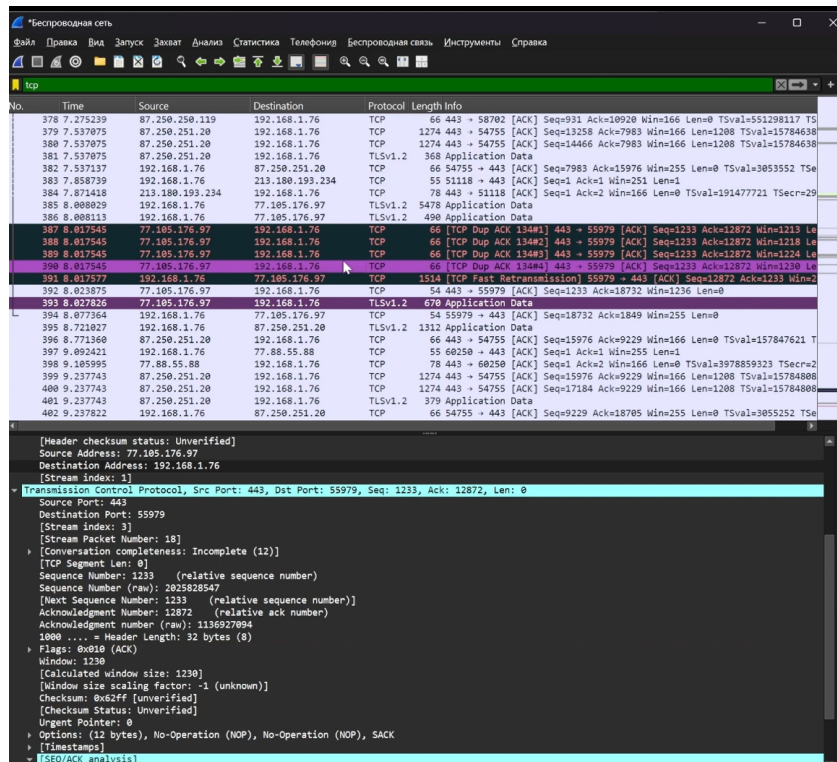


Рис. 2.25: ack

Посмотрим на график потока. На нём хорошо видно хендшейк (SYN, SYN+ACK, ACK) прямо перед запросом GET (рис. 2.26).

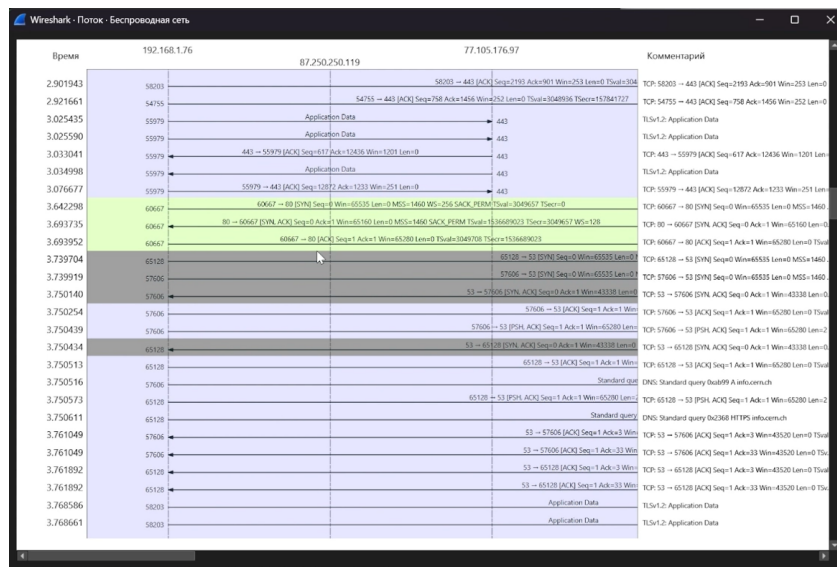


Рис. 2.26: Хендшейк на графике потока

3 Выводы

В результате выполнения работы были получены навыки анализа пакетов в wireshark.

Список литературы

1. Wireshark Foundation. Wireshark User's Guide [Электронный ресурс]. — URL: https://www.wireshark.org/docs/wsug_html_chunked/ (дата обращения: 07.10.2026).
2. Wireshark Foundation. Wireshark Display Filter Reference [Электронный ресурс]. — URL: <https://www.wireshark.org/docs/dfref/> (дата обращения: 07.10.2026).
3. IETF. RFC 826: An Ethernet Address Resolution Protocol [Электронный ресурс]. — URL: <https://datatracker.ietf.org/doc/html/rfc826> (дата обращения: 07.10.2026).